Math 242, S20 **Name:** Answer

Quiz 2

Instructions: There are totally 2 problems. You must show all of your work to receive a credit for a correct answer.

Problem 1. (5 points) Find a general solution (explicit if convenient) for the first-order differential equation $\frac{dy}{dx} = 2x\csc(y)$.

$$\frac{dy}{dx} = 2x\csc(y)$$

$$\frac{dy}{dx} = \frac{2x}{\sin y}$$

$$\sin y\, dy = 2x\, dx$$

$$\int \sin y\, dy = \int 2x\, dx$$

$$-\cos y = x^2 + C$$

$$\cos y = -x^2 - C \Rightarrow y(x) = \arccos(-x^2 - C)$$

Problem 2. (5 points) Find the explicit particular solution for the initial value problem $\frac{dy}{dx} = 2xy - y$, $y(1) = -8$.

$$\frac{dy}{dx} = 2xy - y$$

$$\frac{dy}{dx} = y(2x-1)$$

$$\frac{1}{y}\, dy = (2x-1)\, dx$$

$$\int \frac{1}{y}\, dy = \int 2x - 1\, dx$$

$$\ln|y| = x^2 - x + C_1$$

$$|y| = e^{x^2 - x + C_1}$$

$$= e^{C_1} e^{x^2 - x}$$

$$\Rightarrow y = A e^{x^2 - x} \quad (A = \pm e^{C_1})$$

$$y(1) = -8 \Rightarrow$$

$$-8 = A e^{1^2 - 1}$$

$$A = -8$$

Thus we have

$$y(x) = -8e^{x^2 - x}$$